

3.3 showing that a language is decidable iff some nondeterministic Turing machine decides it.

- ① If a language L is decidable, so L can be decided by a deterministic Turing machine. And a deterministic Turing machine is a nondeterministic Turing machine. So L also can be decided by a nondeterministic Turing machine.
- ② According 3.1.6, if L is decided by some nondeterministic Turing machine with every node in the tree has finitely many children and every branch of the tree has finitely many nodes, the tree itself has finitely many nodes, this is equivalent with deterministic Turing machine. So if nondeterministic Turing machine accepts L , it eventually find a branch to accept this input; else the tree is completely traversed and halts, return REJECT.